



CAT 2025

MBA FASTRACK BATCH

Lecture - 01

Logical DI

Missing Data

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TOPICS *to be covered*



Sets on Missing Data



Set-1

Read the following data and answer the given questions.



#Q. With the spread of Covid-19 in 2020, it was mandated by Public Health Authorities that each patient admitted in the hospital has to undergo Covid-19 test. In the context of this information, study the table a under and answer the given questions.

Number of patients admitted and found COVID-19 positive in a district hospital in 2021.

Months	No. of patients admitted	% of admitted patients found Covid 19 +ve	Respective ratio of number of admitted Covid 19 +ve male and number of admitted Covid 19 +ve female patients.
March	900	<u> </u>	3 : 2
April	<u> </u>	<u> </u>	5 : 3
May	680	60%	<u> </u>
June	<u> </u>	42%	9 : 5
July	1000	64%	<u> </u>

Question-1



#Q. In July 2021, if the number of Covid-19 positive female patients was 194, what was the respective ratio of number of admitted Covid-19 positive male patients and number of admitted Covid-19 positive female patients?

- A. 280: 41^x
- B. 197: 41^x
- ☒ C. 223: 97
- D. 114: 57^x

$$\frac{M}{194} \rightarrow$$

$$640 \rightarrow \begin{cases} F = 194 \\ M = 446 \end{cases}$$

$$\frac{446}{194} = \frac{223}{97}$$

Number of patients admitted and found COVID-19 positive in a district hospital in 2021.

Months	No. of patients admitted	% of admitted patients found Covid 19 +ve	Respective ratio of number of admitted Covid 19 +ve male and number of admitted Covid 19 +ve female patients.
March	900	—	3 : 2
April	—	—	5 : 3
May	680	60%	—
June	—	42%	9 : 5
July	1000	64%	—

Question-2



#Q. If the number of admitted patients increased by 40 percent from March to August 2021 and also, 25 percent of the admitted patients were found Covid-19 positive in August 2021, what was the number of Covid-19 positive patients in August 2021?

A. 240

B. 225

C. 255

☒ D. 315

Aug
 $900 + 40\% \text{ of } 900 = 1260$
 $25\% \text{ of } 1260 = 315$

Number of patients admitted and found COVID-19 positive in a district hospital in 2021.

Months	No. of patients admitted	% of admitted patients found Covid 19 +ve	Respective ratio of number of admitted Covid 19 +ve male and number of admitted Covid 19 +ve female patients.
March	900	—	3 : 2
April	—	—	5 : 3
May	680	60%	—
June	—	42%	9 : 5
July	1000	64%	—

Question-3



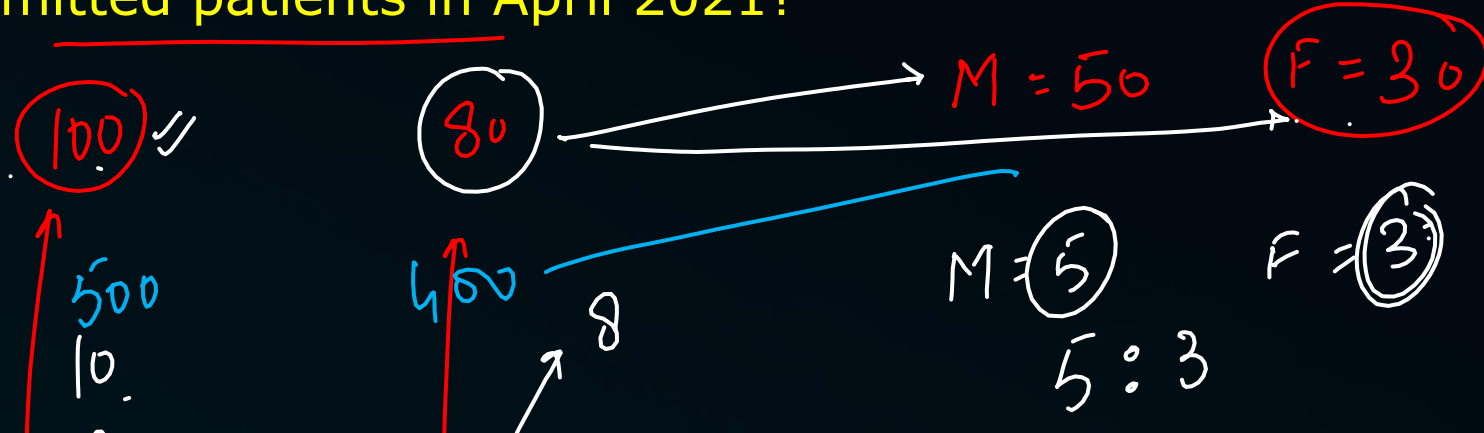
#Q. In April 2021, if the respective ratio of the number of admitted patients to the number of patients found Covid-19 positive was 5:4 then the number of female Covid-19 positive patients constitute what percent of number of admitted patients in April 2021?

A. 30

B. 20 x

C. 25

D. 15



Number of patients admitted and found COVID-19 positive in a district hospital in 2021.

Months	No. of patients admitted	% of admitted patients found Covid 19 +ve	Respective ratio of number of admitted Covid 19 +ve male and number of admitted Covid 19 +ve female patients.
March	900	—	3 : 2
April	—	—	5 : 3
May	680	60%	—
June	—	42%	9 : 5
July	1000	64%	—

Question-4



#Q. In June 2021, if the difference between the number of male Covid-19 positive patients and female Covid-19 positive patients was 72 then what was the number of admitted patients in June 2021?

A. 800

B. 900

C. 850

☒ D. 600

$$100 \times 6 = 600$$

$$100n$$

$$42n$$

$$M = 27n$$

$$F = 15n$$

$$\text{Diff} = 12n = 72$$

$$n = 6$$

Number of patients admitted and found COVID-19 positive in a district hospital in 2021.

Months	No. of patients admitted	% of admitted patients found Covid 19 +ve	Respective ratio of number of admitted Covid 19 +ve male and number of admitted Covid 19 +ve female patients.
March	900		3 : 2
April			5 : 3
May	680	60%	
<u>June</u>		<u>42%</u>	<u>9 : 5</u>
July	1000	64%	



Set-2

Read the following data and answer the given questions.



#Q. Funky Pizzeria was required to supply Pizzas to three different parties. The total number of Pizzas it had to deliver was 800. 70% of which was to be delivered to Party 3 and the rest equally divided between Party 1 and Party 2. Pizzas could be of Thin Crust (T) or Deep Dish (D) variety and come in either Normal Cheese (NC) or Extra Cheese (EC) versions.

Hence, there are 4 types of Pizzas: T – NC,

T – EC, D-NC, D-EC. Partial information

about proportions of T and NC pizzas

ordered by the three parties are given.

	Thin crust (T)	Normal cheese (NC)
Party 1	0.6	
Party 2	0.55	0.3
Party 3		0.65
Total	0.375	0.52

A. 104 B. 84 C. 16 D. 196

A. 18 B. 12 C. 30 D. 24

A. 398 B. 162 C. 196 D. 364

A. Rs.59480 B. Rs 59840 C. Rs 42520 D. Rs 45240

A. 22% **B. 13.5%** **C. 20%** **D. 18.5%**

Analysis

	Total	T	D	NC	EC
P-1	120	72	48	16	104
P-2	120	66	54	36	84
P-3	560	162	398	364	196
Total.	800	300	500	416	384

The total number of Pizzas it had to deliver was 800. 70% of which was to be delivered to Party 3 and the rest equally divided between Party 1 and Party 2. Pizzas could be of Thin Crust (T) or Deep Dish (D) variety and come in either Normal Cheese (NC) or Extra Cheese (EC) versions.

Hence, there are 4 types of Pizzas: T – NC, T – EC, D-NC, D-EC. Partial information about proportions of T and NC pizzas ordered by the three parties are given.

	Thin crust (T)	Normal cheese (NC)
Party 1	0.6	
Party 2	0.55	0.3
Party 3		0.65
Total	0.375	0.52

Questions



#Q. How many Normal Cheese pizzas were required to be delivered to Party 1?

A. 104

B. 84

✓ C. 16

D. 196

	Total	T	D	NC	EC
P-1	120	72	48	16	104
P-2	120	66	54	36	84
P-3	560	162	398	364	196
Total.	800	300	500	416	384

Questions



#Q. For Party 2, if 50% of the Normal Cheese pizzas were of Thin Crust variety, what was the difference between the numbers of T-EC and D-EC pizzas to be delivered to Party 2?

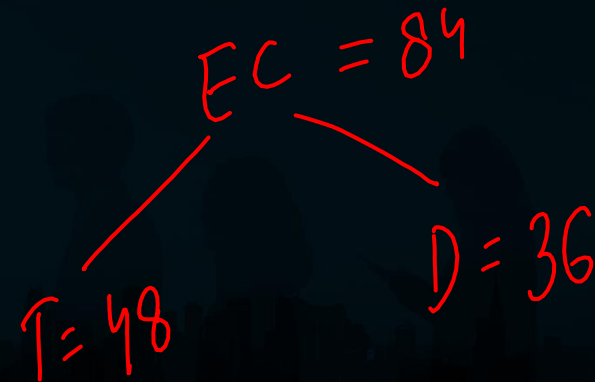
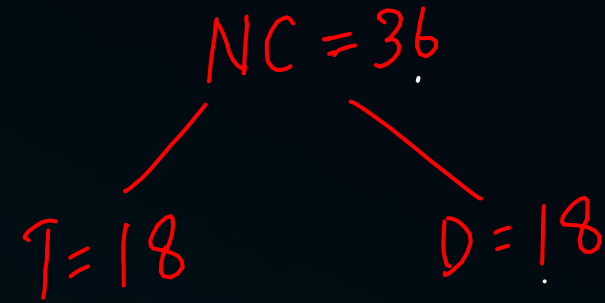
A. 18

✓ B. 12

C. 30

D. 24

	Total	T	D	NC	EC
P-1	120	72	48	16	104
P-2	120	66	54	36	84
P-3	560	162	398	364	196
Total	800	300	500	416	384



$$T - NC = 18$$

$$D - NC = 18$$

$$T - EC = 48$$

$$D - EC = 36$$

$$12$$

Questions



#Q. How many Thin Crust pizzas were to be delivered to Party 3?

A. 398

✓ B. 162

C. 196

D. 364

	Total	T	D	NC	EC
P-1	120	72	48	16	104
P-2	120	66	54	36	84
P-3	560	162	398	364	196
Total.	800	300	500	416	384

Questions



#Q. Suppose that a T-NC pizza cost as much as a D-NC pizza, and both are 60% of the price of a D-EC pizza. A D-EC pizza costs Rs. 50 more than a T-EC pizza, and the latter costs Rs. 500. If 25% of the Normal Cheese pizzas delivered to Party 1 were of Deep-Dish variety, what was the total bill for Party 1?

A. Rs.59480

B. Rs 59840

C. Rs 42520

D. Rs 45240

$$T-EC = 500$$

$$D-EC = 550$$

$$T-NC = 330$$

$$D-NC = 330$$

$$NC = 16 \begin{cases} T = 12 \times 330 = 3960 \\ D = 4 \times 330 = 1320 \end{cases}$$

$$EC = 104 \begin{cases} T = 60 \times 500 = 30,000 \\ D = 44 \times 550 = 24,200 \end{cases}$$

	Total	T	D	NC	EC
P-1	120	72	48	16	104
P-2	120	66	54	36	84
P-3	560	162	398	364	196
Total.	800	300	500	416	384

Questions



#Q. For Party 3, if there are 350 pizzas of the type Deep dish-Normal Cheese (D-NC), then what is the percentage of Thin crust-Extra Cheese (T-EC) ordered in party 3 out of total pizzas ordered in all parties?

- A. 22%
- B. 13.5%
- C. 20%
- ☒ D. 18.5%

$$\frac{148}{800} \times 100$$

$$D = 398 \begin{cases} NC = 350 \\ EC = 48 \end{cases}$$

	Total	T	D	NC	EC
P-1	120	72	48	16	104
P-2	120	66	54	36	84
P-3	560	162	398	364	196
Total	800	300	500	416	384

$$T = 162 \begin{cases} NC = 14 \\ EC = 148 \end{cases}$$



Set-3

Read the following data and answer the given questions.



#Q. A company administers a written test comprising of three sections of 20 marks each: Data Interpretation (DI), Written English (WE) and General Awareness (GA), for recruitment. A composite score for a candidate (out of 80) is calculated by doubling her marks in DI and adding it to the sum of her marks in the other two sections. Candidates who score less than 70% marks in two or more sections are disqualified. From among the rest, the four with the highest composite scores are recruited. If four or less candidates qualify, all who qualify are recruited.

	Marks out of 20		
	DI	WE	GA
Ajay	8		16
Bala		9	11
Chetna	19	4	12
Danish	8	15	
Ester	12	18	16
Falak	15	7	10
Geeta	14		6
Harini	5		
Indu		8	
Jatin		16	14

Some marks in the table are missing, but the following facts are known:

1. No two candidates had the same composite score.
2. Ajay was the unique highest scorer in WE.
3. Among the four recruited, Geeta had the lowest composite score.
4. Indu was recruited.
5. Danish, Harini, and Indu scored the same marks the in GA.
6. Indu and Jatin both scored 100% in exactly one section, and Jatin's composite score was 10 more than Indu's.



- #Q. Which of the following statements must be true?
1. Jatin's composite score was more than that of Danish.
 2. Indu scored less than Chetna in DI.
 3. Jatin scored more than Indu in GA.
- A. Only 2 B. Both 2 and 3 C. Only 1 D. Both 1 and 2

- #Q. Which of the following statements **must** be **false**?
- A. Bala scored the same as Jatin in DI.
- B. Bala's composite score was less than that of Ester.
- C. Chetna scored more than Bala in DI.
- D. Harini's composite score was less than that of Falak.

- #Q. If all the candidates except Ajay and Danish had different marks in DI, and Bala's composite score was less than Chetna's composite score, then what is the maximum marks that Bala could have scored in DI?

- #Q. If all the candidates scored different marks in WE, then what is the maximum marks that Harini could have scored in WE?

Analysis



A company administers a written test comprising of three sections of 20 marks each: Data Interpretation (DI), Written English (WE) and General Awareness (GA), for recruitment. A composite score for a candidate (out of 80) is calculated by doubling her marks in DI and adding it to the sum of her marks in the other two sections. Candidates who score less than 70% marks in two or more sections are disqualified. From among the rest, the four with the highest composite scores are recruited. If four or less candidates qualify, all who qualify are recruited.

	Marks out of 20		
	DI	WE	GA
✓ Ajay	8	19/20	16
✗ Bala	.	9	11
✗ Chetna	19	4	12
Danish	8	15	20
✓ Ester	12	18	16
✗ Falak	15	7	10
✓ Geeta	14	19	6
Harini	5	.	20
✓ Indu	16	8	20
✓ Jatin	20	16	14

CS
~~51/52~~
 54
 51
 58
 47
 53
 60
 70

$$Geeta = 28 + 19 + 6 = 53$$

18 → 52

1. No two candidates had the same composite score.
2. Ajay was the unique highest scorer in WE.
3. Among the four recruited, Geeta had the lowest composite score.
4. Indu was recruited.
5. Danish, Harini, and Indu scored the same marks the in GA.
6. Indu and Jatin both scored 100% in exactly one section, and Jatin's composite score was 10 more than Indu's.

#Q.

Which of the following statements must be true?

1. Jatin's composite score was more than that of Danish. ✓

2. Indu scored less than Chetna in DI. ✓

3. Jatin scored more than Indu in GA. ✗

A. Only 2

B. Both 2 and 3

C. Only 1

D. Both 1 and 2 ✓



	Marks out of 20		
	DI	WE	GA
✓ Ajay	8	19/20	16
✗ Bala	.	9	11
✗ Chetna	19	4	12
Danish	8	15	π 20
✓ Ester	12	18	16
✗ Falak	15	7	10
✓ Geeta	14	19	6
Harini	5	.	π 20
✓ Indu	16	8	π 20
✓ Jatin	20	16	14

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#Q. Which of the following statements **must** be false?

- ✓ A. Bala scored the same as Jatin in DI.
- B. Bala's composite score was less than that of Ester.
- C. Chetna scored more than Bala in DI.
- D. Harini's composite score was less than that of Falak.

	Marks out of 20		
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X Bala	20	9	11
X Chetna	19	4	12
Danish	8	15	20
✓ Ester	12	18	16
X Falak	15	7	10
✓ Geeta	14	19	6
Harini	5	.	20
✓ Indu	16	8	20
✓ Jatin	20	16	14

CS
51/52
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#Q.



If all the candidates except Ajay and Danish had different marks in DI, and Bala's composite score was less than Chetna's composite score, then what is the maximum marks that Bala could have scored in DI?

13

Bala $CS < 54$

$46 = CS$

$2a + 20 < 54$

$a < 17$

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X Bala	a (13)	9	11
X Chetna	19	4	12
Danish	8	15	n 20
✓ Ester	12	18	16
X Falak	15 x	7	10
✓ Geeta	14 x	19	6
Harini	5	.	n 20
✓ Indu	16 x	8	. n 20
✓ Jatin	20	16	14

CS

~~51/52~~

54

51

58

47

53

n 20

n 20

60

70

#Q. If all the candidates scored different marks in WE, then what is the maximum marks that Harini could have scored in WE?



17 X $\rightarrow 5 \times 2 + 17 + 20 = 47$

✓ (14) $\rightarrow 5 \times 2 + 14 + 20 = 44$

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	DI	WE	GA
✓ Ajay	8	19/20	16
X Bala	.	9	11
X Chetna	19	4	12
Danish	8	15	n 20
✓ Ester	12	18	16
X Falak	15	7	10
✓ Geeta	14	19	6
Harini	5	.	n 20
✓ Indu	16	8	n 20
✓ Jatin	20	16	14

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~~51/52~~
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THANK
You

